Development and reflection

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Repository: https://github.com/yerkebttt/municipal-air-quality-batch-ingestion

Implementation and user contracts

I implemented a Python-to-MongoDB batch pipeline around user decisions rather than a generic dashboard. The public-dashboard contract returns the latest 24 station records; analysts receive daily pollutant trends with observation counts; planners compare stations over a chosen policy period; and data stewards receive quality summaries, run status, heartbeats and reconciliation counters. Each query is timed against a configurable 2,000 ms prototype target.

Reliability and data quality

The loader now recognises UCI's nested archive, normalises timezone-aware station-hours, writes 1,000-record batches and safely replays them through deterministic IDs and insert-only upserts. Structural failures enter a quarantine collection. Missing values remain absent; configurable completeness and plausibility checks retain questionable readings with warning flags instead of inventing values. Example thresholds generate review candidates only and cannot notify citizens.

Verification and boundaries

Testing includes eleven unit tests plus a complete source parse. All 420,768 rows were readable: 383,964 passed configured checks and 36,804 carried warnings; none was structurally rejected. Docker Compose provides MongoDB, ingestion and an optional benchmark service. Run `docker compose up --build`, inspect `python -m src.monitoring`, then execute the verification profile. Production still requires municipal sensor metadata, approved health policy, authentication, TLS, backups and measured infrastructure-specific latency.